

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
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Section / Batch:	1	Date:	22/07/2026

Code of Conduct

I _____ Kaviya. K _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Kaviya. K

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic		Focus	Q Nos	Marks
Search Data Structure, Stack		Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure		List Operations	6 - 10	25
Queue Data Structures		Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations		Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks	

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks


10/10 answered

```

}
return -1;
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include<stdio.h>
2 int binarysearch(int arr[],int n,int key){
3     int low =0,high=n-1;
4     while (low<=high){
5         int mid = low +(high - low)/2;
6         if(arr[mid]==key){
7             return mid;
8         }else if (arr[mid]<key){
9             low = mid +1;
10        }else{
11            high = mid -1;
12        }
13    }
14    return -1;
15 }
16

```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```

int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include<stdio.h>
2 int pop() {
3     if (top==0) {
4         printf("underflow\n");
5         return -1;
6     }
7     return stack[top--];
8 }

```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack
5 marks



Q6

Linked List
5 marks



Q7

Stack
5 marks



Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

🌟 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int stack[5];
3 int top;
4 void init() {
5     top=-1;
6 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack
5 marks



Q6

Linked List
5 marks



Q7

Stack
5 marks



Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
    while(stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();
}
```

🌟 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 if(ch==' '){
3     while (stack[top]!='('){
4         postfix[j++] = pop();
5     }
6     pop();
7 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
    pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((ch=='[' && stack[top]=='[') || (ch==']' && stack[top]=='[')) {
2     pop();
3 }else{
4     return 0;
5 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

Q6 Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 node temp=head;
2 while (temp!=null){
3     system.out.println(temp.data);
4     temp=temp.next;
```

Test Input

stdin input (optional)

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {
2     if(top<0) {
3         printf("underflow\n");
4         return -1;
5     }else{
6         return stack[top--];
7     }
8 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head
head = new
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new_node->next = head;
2 head = new_node;
```

Test Input

stdin input (optional)

Output

▶ Run

📁 Save

🔧 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
```

5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

```
int op2 = pop();  
// perform operation: result = op1 + op2;  
push(result);  
}  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 #include<ctype.h>  
3 int evaluatepostfix(char* expression){  
4     int i = 0;  
5     while(expression[i]!='\0'){  
6         if (isdigit(expression[i])){  
7             push(expression[i]-'0');  
8         }else if(expression[i]!=' '){  
9             int op2=pop();  
10            int op1=pop();  
11            switch(expression[i]){  
12                case'+':push(op1 + op2);break;  
13                case'-':push(op1-op2);break;  
14                case'*':push(op1*op2);break;  
15                case'/':push(op1/op2);break;  
16            }  
17            i++;  
18        }  
19    }  
20    return pop();  
21 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;
```

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarysearch(int arr[],int l,int r,int x){
2     if(l>r){
3         return -1;
4     }
5     int mid = l + (r-l)/2;
6     if(arr[mid]==x){
7         return mid;
8     }
9     if(arr[mid]>x){
10        return binarysearch(arr,l,mid-1,x);
11    }
12    return binarysearch(arr,mid+1,r,x);
13 }
```

Test Input

 stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓

Stack Notations - Postfix Expression
5 marks

Q2 ✓

Circular Queue
5 marks

Q3 ✓

Binary Search
5 marks

Q4 ✓

Queue
5 marks

Q5 ✓

Queue
5 marks

Q6 ✓

Doubly Linked List
5 marks

Q7 ✓

Stack
5 marks

Q8 ✓

Stack

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data){
2     if(rear ==MAX -1){
3         printf("Queue overflow\n");
4         return;
5     }
6     if(front ==-1){
7         front =0;
8     }
9     rear++;
10    queue[rear]=data;
11 }
```

Test Input

 stdin input (optional)

Output

 Run

 Save

 Submit Fix

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node**head_ref,node*del) {  
2     if(head_ref==NULL||*head_ref==NULL||del==NULL) {  
3         return;  
4     }  
5     if(*head_ref==del){  
6         *head_ref = del->next;  
7     }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():
if top == 0 → Underflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {  
2     if(top==1){  
3         printf("stack underflow\n");  
4         return -1;  
5     }  
6     int data = stack[top];  
7     top--;  
8     return data;  
9 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void push(int data){
2     if(top==MAX-1){
3         printf("stack overflow\n");
4         return ;
5     }
6     top++;
7     stack[top]=data;
8 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int vehicle){
2     if((rear+1)%MAX==front){
3         printf("queue full\n");
4         return;
5     }
6     if(front == -1){
7         front=0;
8     }
9     rear=(rear+1)%MAX;
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS
Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q10 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int peek() {
2     if(top==1){
3         printf("stack is empty\n");
4         return -1;
5     }
6     return stack[top];
7 }
```

Test Input

stdin input (optional)

Output

▶ Run

☁ Save

🌟 Submit Fix

